

Exercise 7: Multi-Layer Perceptrons - Keras

Task 1:

Below is the implementation of a simple MLP to solve a classification problem on the small given dataset. Find mistakes in the code and explain how you would fix them.

```
import tensorflow as tf
from tensorflow.keras import layers, models
import numpy as np

X = np.array([
    [1.2, 0.7, -3.1, "blue"],
    [0.5, -1.0, 2.2, "red"],
    [-0.2, 4.1, 0.0, "red"],
    [3.3, 1.1, -0.9, "blue"],
    [1.0, 0.0, 1.5, "blue"]
], dtype=object)

y = np.array([0, 2, 1, 0])

model = models.Sequential([
    layers.Dense(16, activation="relu", input_shape=(5,)),
    layers.Dense(8, activation="relu"),
    layers.Dense(3, activation="relu")
])

model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["mae"]
)

model.fit(
    X, y,
    epochs=10,
    batch_size=2
)
```

Task 2:

Check out the *housing_prices.csv* file from Moodle. It contains information about houses, as well as their sale prices. Your task is to train an MLP that predicts the sale prices based on the other features in the dataset. Use all available features and encode/scale them appropriately.

Your MLP should consist of at least one hidden layer.

Use Keras (https://keras.io/guides/functional_api/) for your MLP implementation.

Choose 2 metrics for evaluating your model and explain your choice of metrics.

Train and test your model using a 70/30 split.